Simulations of Interacting Active Brownian Particles in Curved Geometries

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April 30, 2026

What are we doing here?

- Active matter is a thriving field of modern physics research.
- Active matter = nonequilibrium soft matter + self-propulsion.
- Often used to model the behavior of biological systems.
- Active Brownian Particles (ABPs) are individual agents move on a surface with an intrinsic velocity.
- We extend the basic model with Vicsek-style orientational alignment and curved surface geometries.
- The goal: determine if/how curved surfaces effect the behavior of the model.

Basic Active Brownian Particles

The simplest mathematical model of ABPs is given below:

$$\mathbf{r}_i(t + \mathrm{d}t) = \mathbf{r}_i(t) + v_0 \hat{\mathbf{n}}_i \mathrm{d}t$$

$$\theta_i(t + \mathrm{d}t) = \theta_i(t) + \sqrt{2D\mathrm{d}t}\,\xi_i(t)$$

$$i = \{1, 2, \ldots, N\}$$

- $\mathbf{r}_i$: Vector position of particle i. In Euclidean space, $\mathbf{r}_i = \langle x_i, y_i \rangle$
- θ_i : Orientation of particle i.
- $\hat{\mathbf{n}}_i$: Unit heading vector of particle i. In Euclidean, $\hat{\mathbf{n}}_i = \langle \cos\theta_i, \sin\theta_i \rangle$.
- $\xi_i(t)$: Sample from standard normal distribution.
- v_0 : Intrinsic self-propulsion velocity of all particles.
- D : Diffusion coefficient, controls "how random" the motion is.
- N : Number of particles

Vicsek Orientaional Alignment

- In the Vicsek model, particles orient themselves in the average direction of their neighbors.
- Over time, and with a sufficiently large interaction radius, particles “flock” together.
- Modelled as an additional term in the time-step for θ:

$$\theta_{Vicsek,i}(t + \mathrm{d}t) = C \sin(\theta_{mean} - \theta_i)\mathrm{d}t$$

$$\theta_{\mathrm{mean}} = \mathrm{atan2}\left(\sum_{j \neq i} \sin\theta_j, \sum_{j \neq i} \cos\theta_j\right)$$

- C is the *coupling strength* parameter.

Differential Geometry for Dummies

- Curved surfaces may be generalized with the concept of *manifolds*.
- General manifolds are complicated objects, but we are concerned with smooth (differentiable) two-dimensional Riemannian manifolds (embedded in $\mathbb{R}^3$).
- Every point on a differentiable manifold has a local tangent space, allowing us to define velocity vectors (our θ_i are angles measured on the tangent space).
- Riemannian manifolds are equipped with extra structure known as a *metric* g, which allows the measurement of distances.
- The metric is defined with two indices: g_{ij}, and for two-dimensional manifolds is given as a 2×2 tensor.
- A manifold may be given a coordinate mapping, which in general in two-dimensions we'll call q_1 and q_2.

Differential Geometry the Lightly Educated

- Tangent spaces rotate due to curvature across the surface.
- Without correcting for this, our particles experience a "phantom torque" following the surface curvature.
- We add another term to θ, called the *spin connection*.

$$\omega_\mu^{ab} = e_\nu^a \Gamma^\nu_{\sigma\mu} e^{\sigma b} - e^{\nu b} \partial_\mu e_\nu^\alpha$$

$$\theta_{\text{spin},i}(t + \mathrm{d}t) = \omega_\mu^{ab} \mathrm{d}q^\mu$$

- e_ν^a are *vierbein* defined implicitly w.r.t. the *metric* through $g_{ij} = e_i^a \delta_{ab} e_j^b$. They allow us to translate between the coordinates on the manifold and the local tangent space.
- Γ^i_{jk} are the *Christoffel symbols*, also defined with respect to the metric, $\Gamma^i_{jk} = \frac{1}{2} g^{ii} \left(\partial_j g_{ik} + \partial_k g_{ij} - \partial_i g_{jk}\right)$. They measure misalignment between straight-line vs curved-surface motion.

Further Consequences of Differential Geometry

- Our Vicsek update involves checking which other particles are nearby, within some interaction radius.
- The Euclidean distance, or the Pythagorean theorem, $d(\mathbf{a}, \mathbf{b}) = \sqrt{\mathrm{d}q_1^2 + \mathrm{d}q_2^2}$ (where $\mathrm{d}q_1 \approx q_{1,a} - q_{1,b}$ and $\mathrm{d}q_2 \approx q_{1,a} - q_{1,b}$) doesn't accurately measure distance on curved manifold.
- We need the geodesic distance, which with a diagonal metric is approximately:

$$d(\mathbf{a}, \mathbf{b}) = \sqrt{g_{11}\mathrm{d}q_1^2 + q_{22}\mathrm{d}q_2^2}$$

- We also can't directly average angles in the Vicsek angles, we first need to *parallel transport* the tangent spaces at the two points so the angles mean the same thing.

Simulation Setup

- The core of the majority of our simulation is built in C++
- C++ is generally faster than Python, particularly for many nested loops of numerics.
- We make use of modular header files to keep code organized.
- Data is analyzed and visualized in Python, since ecosystems are both more robust and easier to work with there.
- Parameter sweeps over D and C are performed in parallel with a bash script calling the compiled C++ executable.

```
├── cpp
│   ├── include
│   │   ├── dynamics.h
│   │   ├── h5sim.h
│   │   ├── h5write.h
│   │   ├── manifold.h
│   │   ├── state.h
│   │   └── vec.h
│   ├── src
│   │   ├── core.cpp
│   │   ├── dynamics.cpp
│   │   ├── h5sim.cpp
│   │   ├── manifold.cpp
│   │   ├── state.cpp
│   │   └── vec.cpp
│   └── CMakeLists.txt
├── python
│   ├── analysis.ipynb
│   └── animate.py
├── scripts
│   └── exp.sh
└── .gitignore
```

Specific Manifolds

- I chose to implement a spherical and toroidal manifold, as well as regular Euclidean space as a control.
- The coordinate chart on the sphere is as usual: $q_1 = \phi \in [0, 2\pi)$ (azimuthal angle), $q_2 = \alpha \in [0, \pi)$ (polar angle, usually θ but that's taken). For radius R, the metric is:

$$g_{\text{sphere}} = \begin{bmatrix} R^2 \sin^2 \alpha & 0 \\ 0 & R^2 \end{bmatrix}$$

- The coordinate chart on the sphere is $q_1 = \phi \in [0, 2\pi)$ (toroidal angle) $q_2 = \psi \in [0, 2\pi)$ (poloidal angle). With major radius R and minor radius r, the metric is:

$$g_{\text{torus}} = \begin{bmatrix} (R + r\cos\psi)^2 & 0 \\ 0 & r^2 \end{bmatrix}$$

The Vaguest Possible Pseudocode Overview

procedure $\textsc{StepSimulation}(\Delta t)$

  Phase 1: Geometry & Dynamics

  for each particle i **do**

    $\theta_i \leftarrow \theta_i + \omega + \sqrt{2D\Delta t} \cdot \mathcal{N}(0,1)$ ▷ Update Orientation

    $q_i \leftarrow q_i + \text{Vielbein}(q_i) \cdot \vec{v}(\theta_i)\Delta t$ ▷ Update Position

    $\text{Manifold.wrap}(q_i)$ ▷ Boundary conditions

  end for

  Phase 2: Interaction & Alignment

  for each pair (i,j) within R_{int} **do**

    if $d(i,j) < 2s_{radius}$ **then**

      $\text{ResolveCollision}(i,j)$ ▷ Elastic reflection

    end if

    $\text{ParallelTransport}(\theta_i \leftrightarrow \theta_j)$ ▷ Align tangent spaces

    $\theta_i \leftarrow \theta_i + \sum \sin(\theta_{\text{mean}} - \theta_i)\Delta t$ ▷ Vicsek alignment

  end for

end procedure

Investigation of the Scientific Question

- At the beginning, I said the goal was to see how different manifolds exhibit different behavior.
- The specific behavior we're looking at is the phase-transition that comes from the interplay of the diffusion and coupling coefficients.
- At high diffusion, the particles act like a gas, moving randomly and overcoming their orientational alignment.
- At high coupling, the particles act like a fluid, flowing in groups along the surface.
- Each "experiment" run consists of running the algorithm on the previous slide for a large number of timesteps on each of our three manifolds, with four different values of C and D.
- By computing the global order parameter for each manifold over different sets of parameters, we can see how alignment behaves as a function of C, D, and the manifold.
- For fun, we also animate the particle trajectories.

Global Order Parameter

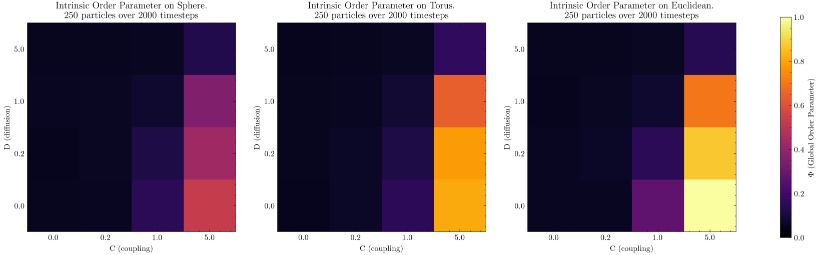
- To compare the alignment behavior across manifolds, we modify the typical Vicsek global order parameter:

$$\Phi = \left\langle \left| \frac{1}{N} \sum_{i}^{N} \mathbf{v}_i \right| \right\rangle$$

- It ranges from 0 to 1, with 0 being complete random alignment (gas-like behavior) and 1 being complete global alignment (flocking) of all particles.
- $\langle \cdot \rangle$ is a time average over the end of the simulation, ensuring steady-state behavior rather than a transient coincidence.
- Particles moving in the same direction on a curved surface will not have the same $\mathbf{v}_i$ vector in 3D space; we must first parallel transport the vectors to the same location on the surface, with $\bar{\theta}_i$ equal to the orientation after transport.

$$\Phi_{\text{intrinsic}} = \left\langle \left| \frac{1}{N} \sum_{i}^{N} e^{i\bar{\theta}_i} \right| \right\rangle$$

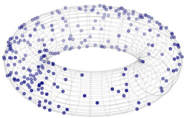
Global Order Parameter Results

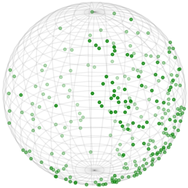


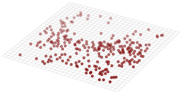
- In general, we see what we expect: particles align far more in low diffusion, high coupling regimes.
- The phase transition is likely gradual; our parameter sweep is too restricted to see intermediate behavior.
- In general, curvature makes global alignment more difficult.

Animations?

Embedding animations in a PDF is a pain. I've gathered some still preview images for completeness. Pivoting to live demonstration!







In Conclusion

- Successfully implemented a C++/Python pipeline for simulating interacting ABPs on arbitrary 2D Riemannian manifolds.
- Quantitatively demonstrated phase-transition across parameter sweeps for different manifolds.
- In general, it seems it is more difficult for particles to align on curved surfaces.
- Further testing with larger parameter sweeps, investigations into initial particle density, etc, will be necessary to codify this result.

Thank You

Questions?